

Agricultural Restructuring in Mainland Portugal

Regional and Temporal Patterns from 1989 to 2019



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1. Introduction

Agriculture remains an important dimension for understanding territorial and economic change in Portugal. Although its relative weight in the national economy has declined over time, it continues to shape rural landscapes, employment structures and land-use patterns. Therefore, studying agricultural change is not only about production, but also about understanding how rural territories have been transformed.

In recent decades, Portuguese agriculture has undergone significant structural changes. The 2019

Agricultural Census shows a decrease in the number of agricultural holdings compared with 2009, while the utilized agricultural area increased, suggesting a process of concentration and restructuring rather than a simple decline of the sector (INE, 2020). This transformation is relevant because it affects how agricultural activity is organized, who works in agriculture and how land is used across different regions.

Agricultural labour is one of the clearest indicators of this process. Its decline may reflect broader changes such as mechanization, ageing rural populations and the weakening of small-scale family farming. At the same time, changes in temporary crops, permanent crops, grasslands and productivity help reveal how agricultural systems have evolved differently across mainland Portugal. These dimensions are also commonly used in European agricultural statistics to analyze farm structure and agricultural transformation (Eurostat, 2024).

This project analyses the structural transformation of agriculture in mainland Portugal between 1989 and 2019, using data from the INE database. The main objective is to understand how agricultural labour, land use, crop systems and productivity evolved over time, and whether these changes suggest decline, concentration, regional specialization or a combination of these processes.

The main research question guiding this project is:

How did the structure of agriculture in mainland Portugal change between 1989 and 2019?

To answer this question, the analysis is guided by the following hypotheses:

H1: Agricultural labour decreased between 1989 and 2019, reflecting a long-term reduction in the labour force involved in agricultural activities.

H2: Agricultural land use changed over time, with different trajectories for temporary crops, permanent crops, grasslands and pastures.

H3: Temporary crops and permanent crops did not follow the same pattern of change, suggesting different forms of agricultural restructuring.

H4: Permanent grasslands and pastures became an important indicator of land-use transformation, especially in regions more associated with extensive agricultural systems.

H5: Productivity patterns varied across agricultural activities and regions, indicating that agricultural change was not homogeneous throughout mainland Portugal.

This overall pattern is summarized in Figure 1, which shows the structural change in mainland Portuguese agriculture between 1989 and 2019. Agricultural labour, temporary-crop holdings, permanent-crop holdings, and agricultural area all declined over time, while permanent grasslands expanded. The figure provides a compact visual overview of the restructuring process examined in this project and helps anticipate the main trends discussed in the following sections.

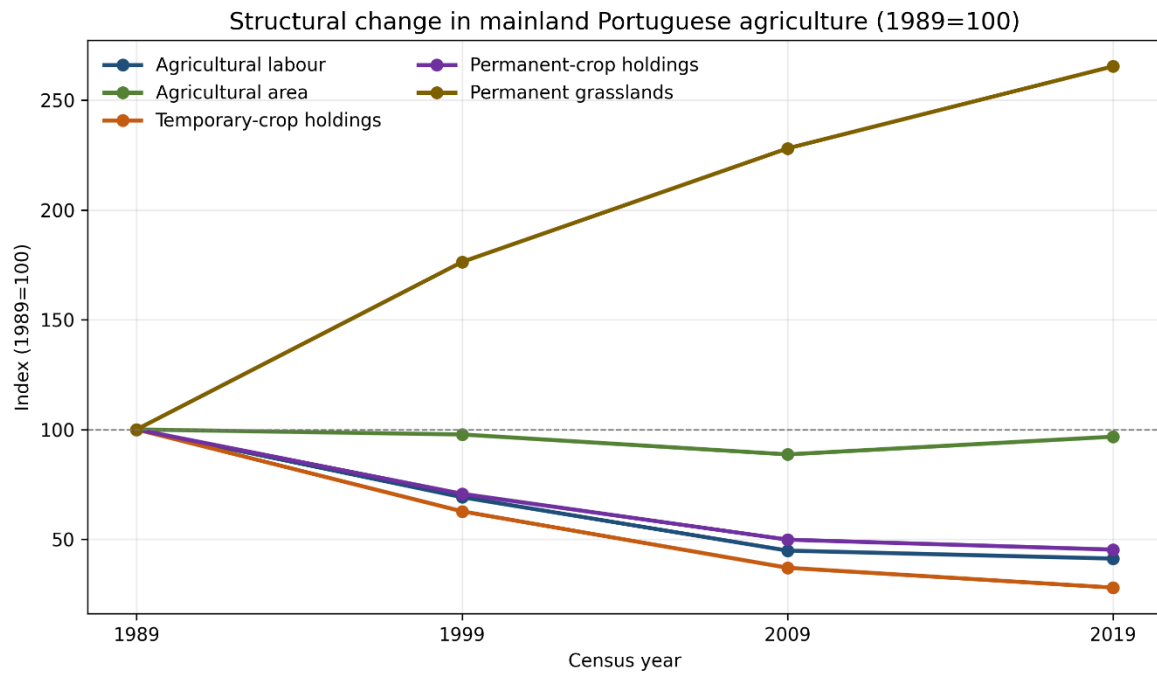


Figure 1: Chart showing Structural change in mainland Portuguese agriculture (1989 – 2019)
Source: INE - Values for “Continente”.

Overall, this project does not approach Portuguese agriculture only as a declining sector. Instead, it understands agricultural change as a process of restructuring, in which labour, land use, crop systems and productivity evolve unevenly across time and space.

2. Database Description

The data used in this project were collected from the Statistics Portugal database and focus on agricultural indicators for mainland Portugal between 1989 and 2019. The datasets cover different dimensions of agricultural structure, including agricultural labour, utilized agricultural area, temporary crops, permanent crops, permanent grasslands and pastures, and productivity of main agricultural crops. These variables were selected because they allow the analysis of agricultural transformation from different perspectives: labour, land use, crop systems and productivity.

2.1 Description of the datasets

The project uses several datasets related to different dimensions of agricultural activity in mainland Portugal. Each dataset focuses on a specific aspect of agricultural structure, allowing the analysis to combine information on labour, land use, crops and productivity.

The agricultural labour dataset is used to examine changes in the number of people working in agriculture. The study area dataset provides information on agricultural land, making it possible to analyze changes in the spatial dimension of agricultural activity. The datasets on temporary crops and permanent crops allow a comparison between different types of crop systems, while the dataset on permanent grasslands and pastures is used to identify changes in extensive land use. Finally, the productivity dataset provides information on the productive performance of main agricultural crops over time.

Together, these datasets make it possible to analyze Portuguese agriculture as a multidimensional process of transformation, rather than focusing only on one indicator.

Description of the datasets used in the analysis			
Dataset	Main indicator	What it measures	Use in the project
Agricultural labour	Agricultural labour force	Number of people working in agriculture	To analyse whether agricultural labour decreased between 1989 and 2019
Study area	Agricultural area	Area used for agricultural activity	To observe changes in land use across mainland Portugal
Temporary crops	Temporary crop holdings / area	Crops cultivated for a limited production cycle	To analyse changes in temporary crop systems
Permanent crops	Permanent crop holdings / area	Crops that remain in the soil for several years	To compare long-term crop systems with temporary crops
Permanent grasslands and pastures	Grasslands and pastures area	Area occupied by permanent grasslands and pastures	To identify changes in extensive land use and land management
Productivity of main agricultural crops	Crop productivity	Productive performance of selected agricultural crops	To analyse whether structural changes were accompanied by changes in productivity

Table 1: Description of the datasets used in the analysis.

This table presents the main datasets used in the project, including their key indicators, measurement focus and analytical purpose. Together, these datasets support the analysis of agricultural labour, land use, crop systems and productivity in mainland Portugal between 1989 and 2019.

2.2 Variables used in the analysis

The main variables used in the analysis are year, geographical area, agricultural labour, agricultural area, crop type, grasslands and pastures, and productivity values. The year variable is essential for identifying changes over time, while the geographical variable allows the analysis of regional differences within mainland Portugal.

Agricultural labour is used to assess whether the number of people working in agriculture decreased between 1989 and 2019. Land-use variables are used to examine how agricultural areas, temporary crops, permanent crops, and grasslands evolved during the period under analysis. Productivity indicators are included to determine whether changes in agricultural structure were accompanied by changes in productive performance.

These variables were selected because they are directly connected to the research question and to the hypotheses defined in the introduction.

Variables used in the analysis		
Variable	Meaning	Purpose in the analysis
Year	Reference year of each observation	To analyse agricultural evolution over time, especially between 1989 and 2019
Region / geographical area	Territorial unit of each observation	To compare regional patterns within mainland Portugal
Agricultural labour	Labour force involved in agricultural activity	To test whether agricultural labour declined over the period analysed
Agricultural area	Area used for agricultural activity	To observe changes in land use and agricultural concentration
Temporary crops	Crops cultivated for a limited production cycle	To examine the evolution of temporary crop systems
Permanent crops	Crops that remain in the soil for several years	To compare long-term crop systems with temporary crops
Permanent grasslands and pastures	Area occupied by grasslands and pastures	To analyse changes in extensive land use and land management
Productivity	Productive performance of selected agricultural crops	To assess whether structural change was accompanied by changes in productivity

Table 2: Variables used in the analysis.

This table presents the main variables used in the project, explaining their meaning and their analytical purpose in the study of structural change in mainland Portuguese agriculture between 1989 and 2019.

2.3 Data preparation and cleaning

Before starting the analysis, the datasets were imported into Python using the pandas library. Each file was inspected to understand its structure, identify the available variables, check the data types, and detect possible missing values.

The first step consisted of opening each CSV file and examining the first rows of the data. Then, the column names were checked and standardized to make the datasets easier to work with. The presence of missing values was also verified, together with the temporal coverage and the geographical units available in each dataset.

After this initial inspection, the relevant variables were selected according to the objectives of the project. When necessary, the data were grouped by year and region to create summary tables for the exploratory analysis. These prepared datasets were then used to produce descriptive statistics, visualizations, and inferential tests.

The cleaning process also ensured that the processed datasets preserved missing values correctly and maintained consistency across the different agricultural indicators used in the analysis.

3. Exploratory Data Analysis

This section presents the first analytical reading of the data through descriptive statistics and visualizations. Its purpose is to identify the main temporal and regional patterns before the inferential analysis. The analysis focuses on the indicators directly connected to the research hypotheses: agricultural labour, agricultural area, temporary crops, permanent crops, permanent grasslands and pastures, and productivity.

Together, these dimensions make it possible to assess whether agricultural change in mainland Portugal followed patterns of decline, restructuring, or regional differentiation. The results presented in this section provide the basis for the statistical tests and for the final interpretation of the hypotheses.

3.1. Evolution of Agricultural Labour in Mainland Portugal

According to INE data, agricultural labour in mainland Portugal decreased substantially between 1989 and 2019. The table below shows the evolution of the total agricultural workforce in 1989, 1999, 2009 and 2019.

Year	Agricultural Labour
2019	596.938
2009	649.607
1999	1.002.480
1989	1.449.906

Table 3: Agricultural labour in mainland Portugal (1989 – 2019)

Source: INE - Agricultural labour by geographic location (NUTS 2013), type of labour and work duration regime. Values for “Continente”.

From this table, approximately 1.45 million people were employed in agriculture in mainland Portugal in 1989. By 2019, this number had fallen to just under 597,000, representing a reduction of approximately 59%. Figure 2 illustrates this evolution more clearly.

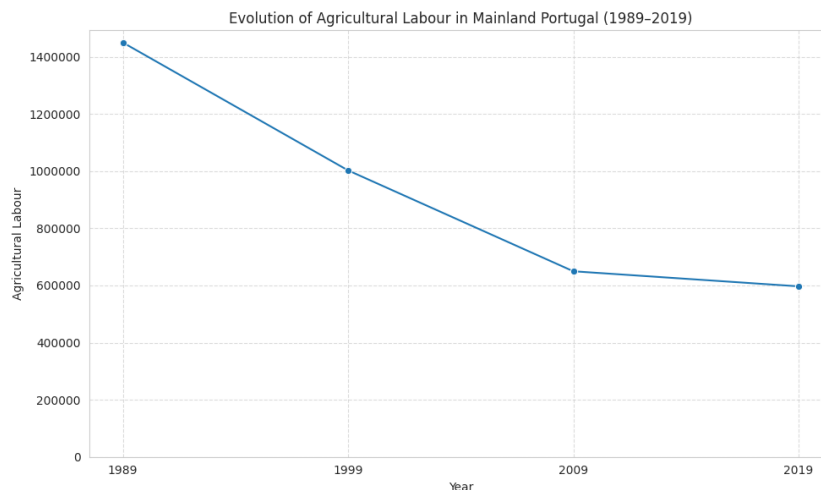


Figure 2: Chart showing the Evolution of agricultural labour in mainland Portugal (1989 – 2019)

Source: INE - Agricultural labour by geographic location (NUTS 2013), type of labour and work duration regime. Values for “Continente”.

Three key observations stand out:

1. Steep initial decline (1989–1999): the loss of nearly 450,000 workers in the first decade reflects a period of intense rural exodus, mechanization, and the abandonment of small, non-viable farms following Portugal’s integration into the European Economic Community in 1986.
2. Continued but slower decline (1999–2009): the reduction of another 350,000 workers suggests that restructuring continued, although at a more moderate pace as the agricultural sector adapted to EU policies, including the Common Agricultural Policy.

3. Moderation of decline (2009–2019): the smaller reduction of 52,669 workers suggests that the agricultural labour force may be approaching a lower baseline. This remaining workforce likely consists of more specialized, full-time, and professional farmers.

Overall, these patterns are consistent with H1 and with a process of agricultural restructuring, in which fewer workers are needed to manage larger, more mechanized, and more capital-intensive farms. The slower decline after 2009 may also reflect a certain stabilization of the sector after decades of transformation.

3.2. Changes in agricultural area / land use

Beyond the decline in agricultural labour, understanding how agricultural land use evolved is essential for characterizing the restructuring process. This section examines changes in total agricultural areas, temporary crop holdings, permanent crop holdings, and the relationship between these different land use categories between 1989 and 2019.

Figure 3 shows agricultural area per agricultural worker, highlighting a clear increase over time. This indicates that fewer workers were managing a relatively similar amount of land, supporting the interpretation of concentration and labour-saving restructuring in mainland Portuguese agriculture.

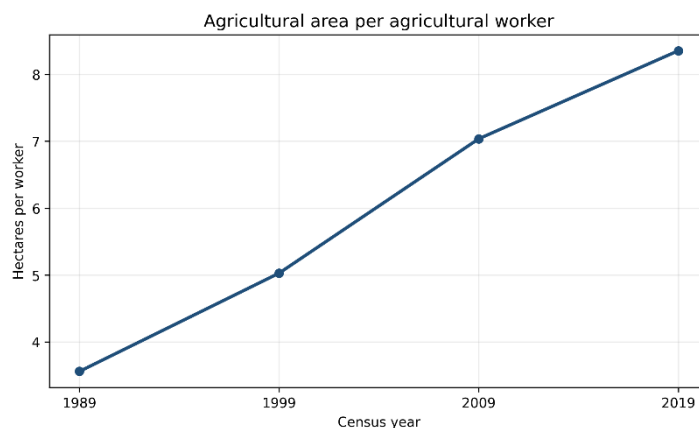


Figure 3: Chart showing the Agricultural area per agricultural worker in mainland Portugal (1989 – 2019)
Source: INE - Agricultural labour by geographic location (NUTS 2013), type of labour and work duration regime AND Agricultural holdings area by geographic location (NUTS 2013) Values for "Continente".

3.2.1 Evolution of Total Agricultural Area

Contrary to the sharp decline observed in agricultural labour, total agricultural area in mainland Portugal remained relatively stable over the 30-year period. Table 3.2 presents the evolution of the total agricultural holdings area between 1989 and 2019.

Year	Agricultural Holdings Area (ha)
1989	5.157.213
1999	5.039.569
2009	4.571.531
2019	4.987.658

Table 4: Total Agricultural Holdings Area in Mainland Portugal (1989 - 2019)
Source: INE – Agricultural holdings area by geographic location (NUTS 2013). Values for "Continente".

The total agricultural area decreased by only 3.3% between 1989 and 2019, from approximately 5.16 million hectares to 4.99 million hectares. This stands in stark contrast to the 59% reduction in agricultural labour over the same period, suggesting that land abandonment was not the primary driver of labour decline.

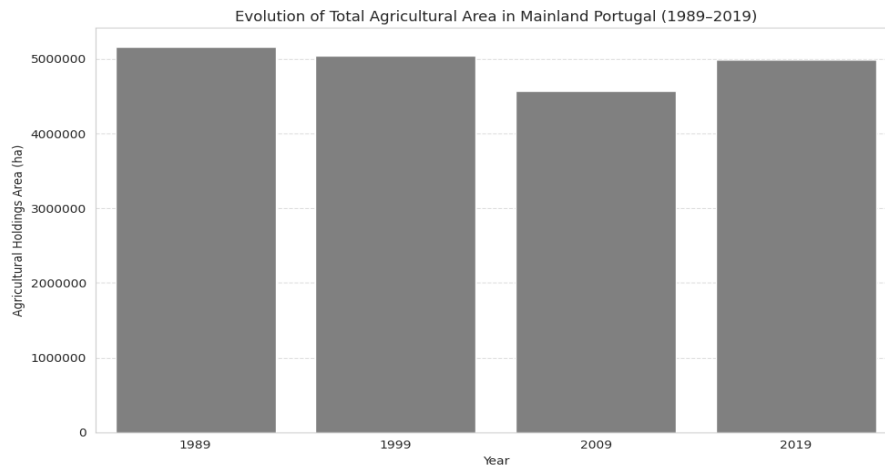


Figure 4: Chart showing the Evolution of Total Agricultural Area (1989–2019)

Source: INE – Agricultural Area in Mainland Portugal (NUTS 2013). Values for "Continente".

Three key observations emerge from this analysis:

1. **Relative stability (1989-1999):** The agricultural area remained nearly constant between 1989 and 1999, decreasing by only 117.644 ha (2.3%). This suggests that during the first decade of analysis, labour decline was driven primarily by mechanization, and efficiency gains rather than land abandonment.
2. **Notable decline (1999-2009):** The period between 1999 and 2009 saw the largest reduction in agricultural area, with a loss of 468.038 ha (9.3%). This coincides with the continued adjustment to CAP reforms and may reflect the abandonment of marginal agricultural land, particularly in less productive regions.
3. **Recovery and stabilization (2009-2019):** Between 2009 and 2019, agricultural area increased by 416.127 ha (9.1%), nearly recovering to 1999 levels. This recovery may be associated with the expansion of permanent crops, particularly olive groves and vineyards, which have been supported by EU investment programs (INE, 2020).

These results suggest that land-use change was not driven by a uniform reduction in agricultural land, but by a reorganization of how land was being used

3.2.2 Land Use Concentration and Restructuring

The data reveals the process of agricultural concentration and restructuring. While total agricultural area decreased only slightly and the number of holdings declined much more sharply, this suggests that farms became larger on average. This is consistent with the restructuring hypothesis: agricultural land is being managed by fewer, larger, and more specialized operations.

Year	Average Temporary Holding Size (ha)	Average Permanent Holding Size (ha)
1989	11.5	11.4
1999	18	15.7
2009	27.7	20.2
2019	39.9	24.3

Table 5. Average Holding Size by Crop Type (1989 - 2019)

Source: INE – Agricultural Area in Mainland Portugal (NUTS 2013). Values for "Continente".

The average temporary crop holding size increased by 247% (from 11.5 to 39.9 ha), while average permanent crop holding size increased by 113% (from 11.4 to 24.3 ha). This suggests that the consolidation process was more pronounced in temporary cropping systems, where mechanization and scale economies are more readily achievable.

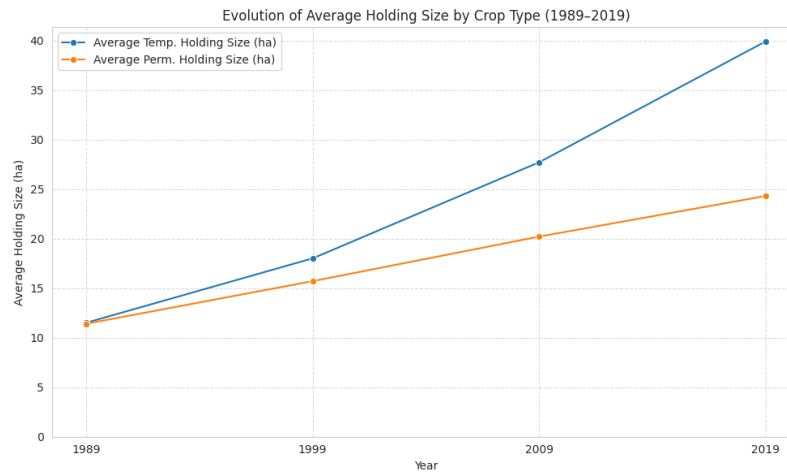


Figure 5: Average Holding Size by Crop Type (1989 - 2019)

Source: INE Agricultural area by geographic location (NUTS 2013), type of crop (temporary and permanent). Values for "Continente".

The sharp increase in average holding size, combined with the reduction in labour intensity, points to a clear pattern of agricultural intensification: fewer, larger farms employing less labour per hectare, particularly in temporary cropping systems. Taken together, these results are consistent with H2, confirming that agricultural land use changed over time and followed different trajectories across crop systems. The increase in average holding size points to a process of concentration and restructuring, particularly in temporary crops

3.3. Temporary and permanent crops

The distinction between temporary and permanent crops is fundamental for understanding agricultural restructuring in mainland Portugal. Temporary crops are cultivated on a seasonal basis and require annual replanting (cereals, pulses, potatoes, horticultural crops), while permanent crops remain in the ground for multiple years (vineyards, olive groves, fruit trees). These two categories respond differently to economic incentives, policy frameworks, and environmental conditions, making their comparative analysis particularly revealing.

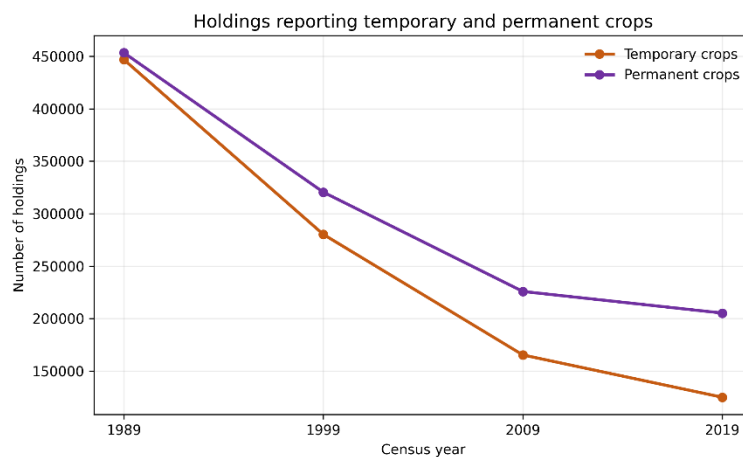


Figure 6: Holdings reporting temporary and permanent crops (1989–2019).

Source: INE – Agricultural area in mainland Portugal (NUTS 2013). Values for "Continente".

Figure 6 shows that both temporary and permanent crop holdings declined between 1989 and 2019, but the decrease was stronger for temporary crops. This suggests that the restructuring process affected the two systems differently, with temporary crops experiencing a faster contraction. Taken together, these results are consistent with H3, indicating that temporary and permanent crops followed different trajectories over time.

3.3.1 Evolution of Temporary Crop Holdings

Temporary crops experienced a sharp decline among all agricultural indicators analyzed. Table 6 presents the evolution of temporary crop holdings by category between 1989 and 2019.

Crop Category	1989	1999	2009	2019	Change (%)
Total Temporary Crops	446.915	280.214	165.265	124.902	-72,1
Grain cereals	352.158	190.446	108.524	66.584	-81,1
Dried pulses for grain	222.230	87.021	28.882	15.945	-92,8
Temporary meadows	40.645	19.531	13.293	14.066	-65,4
Forage crops	285.468	180.000	102.983	65.416	-77,1
Potatoes	312.443	163.360	56.969	34.570	-88,9
Sugar beet	33	495	-	-	-
Industrial crops	3.638	3.738	1.364	1.147	-68,5
Horticultural crops	77.052	48.610	18.141	15.874	-79,4
Flowers and ornamental plants	1.788	1.744	1.056	1.005	-43,8
Other temporary crops	65.133	20.662	1.424	1.718	-97,4

Table 6: Temporary Crop Holdings by Category in Mainland Portugal (1989–2019)

Source: INE – Temporary crops holdings by geographic location (NUTS 2013). Values for "Continente"

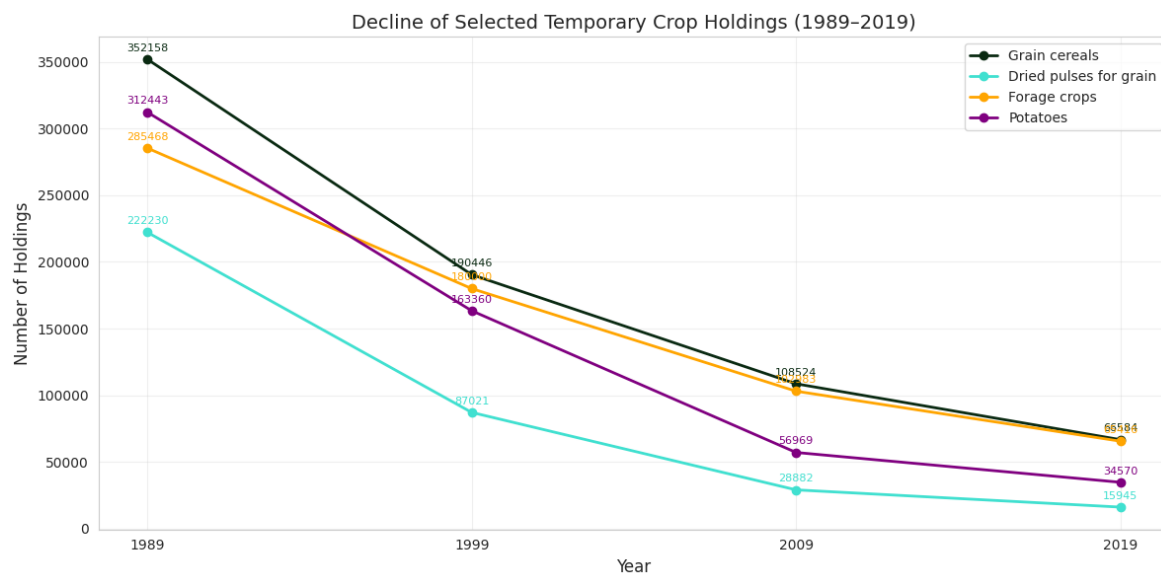


Figure 7: Evolution of Major Temporary Crop Holdings (1989–2019)

Source: INE – Temporary crops holdings by geographic location (NUTS 2013). Values for "Continente"

The data reveals an extraordinary contraction in temporary crop holdings, with the total declining by 72,1% between 1989 and 2019. However, the decline was not uniform across categories. Several patterns stand out:

Decline in traditional arable crops.

- **Dried pulses for grain** experienced the most severe decline (-92,8%), from 222.230 to 15.945 holdings. This reflects the abandonment of traditional legume cultivation in favor of more profitable crops or grassland conversion.
- **Potatoes** declined by 88,9% (from 312.443 to 34.570 holdings), indicating a strong concentration of production in fewer, more specialized operations, particularly in the Centre and North regions.
- **Grain cereals** declined by 81,1% (from 352.158 to 66.584 holdings), consistent with the consolidation of cereal production in the Alentejo and Ribatejo regions.

Decline in forage and temporary meadow systems.

- **Forage crops** declined by 77,1% (from 285.468 to 65.416 holdings), while **temporary meadows** declined by 65,4% (from 40.645 to 14.066 holdings). This is significant because forage crops are closely linked to livestock production. Their decline suggests a shift away from crop-based animal feed toward more extensive grazing systems on permanent grasslands

Decline in Horticultural crops.

- Horticultural crops declined by 79,4% (from 77.052 to 15.874 holdings). This may reflect the concentration of horticultural production in specialized regions, such as the Algarve for vegetables and the Oeste for fruit, together with the consolidation of small holdings into larger operations.

Strong contraction in residual category.

- Other temporary crops declined by 97,4% (from 65.133 to 1.718 holdings), suggesting that many small-scale and diversified farms disappeared over the period.

3.3.2 Evolution of Permanent Crop Holdings

While temporary crop holdings declined sharply, permanent crop holdings followed a different trajectory, as shown in Table 7:

Year	Total Permanent Holdings	Change from Previous Period	Cumulative Change
1989	453.441	-	-
1999	320.484	-0,293	-0,293
2009	225.806	-0,295	-0,589
2019	205.208	-0,091	-0,68

Table 7: Permanent Crop Holdings (1989 – 2019)

Source: INE – Permanent crops holdings by geographic location (NUTS 2013). Values for "Continente".

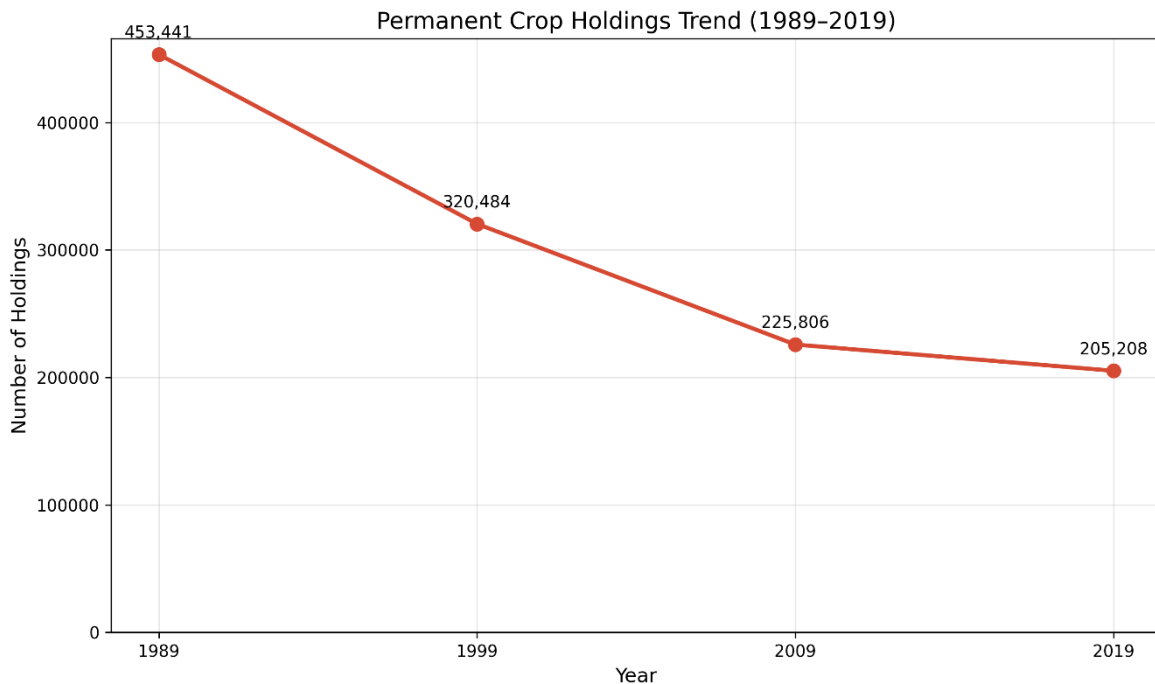


Figure 8: Chart showing Evolution of Permanent Crop Holdings Trends (1989–2019)

Source: INE – Permanent crops holdings by geographic location (NUTS 2013). Values for "Continente".

While temporary crop holdings declined sharply, permanent crop holdings followed a different trajectory. The total number of permanent holdings fell from 453.441 in 1989 to 205.208 in 2019, representing a decline of 54,8%. This decrease was smaller than that observed in temporary crops, and the pace of contraction slowed noticeably after 2009, with only a 9,1% reduction between 2009 and 2019. This suggests that permanent crop holdings may have been approaching a more stable level by the end of the period. This is likely reflected:

- The expansion of olive groves (super-intensive systems) and vineyards, which increased demand for permanent crop holdings;
- The conversion of marginal arable land to permanent crops;
- The professionalization of permanent crop farming, with larger, more viable operations.

Taken together, these results are consistent with H3, indicating that permanent crops followed a different and more resilient trajectory than temporary crops.

3.3.3 Comparative Analysis: Temporary vs. Permanent Crop Holdings (1989 - 2019)

The divergent trajectories of temporary and permanent crops reveal fundamental differences in the restructuring process. Figure 9 summarizes these contrasting patterns.

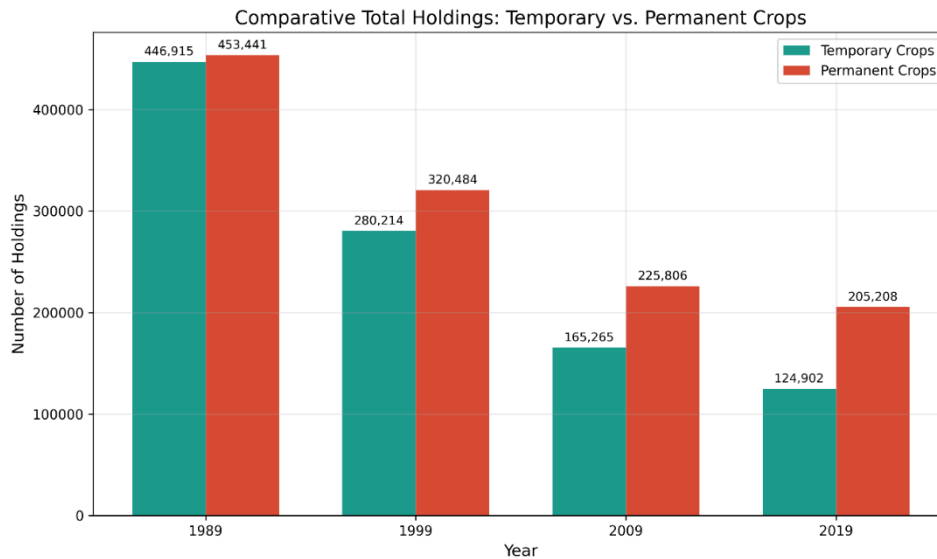


Figure 9: Chart showing Comparative Analysis: Temporary vs. Permanent Crop Holdings (1989–2019)

Source: INE – Permanent crops holdings by geographic location (NUTS 2013). Values for "Continente" and Temporary crops holdings by geographic location (NUTS 2013). Values for "Continente"

Key Observations:

1. Diverging trajectories: While both types of crop holdings declined, permanent crops declined less and showed stronger signs of stabilization after 2009. This suggests that permanent crop systems may have been more resilient to the pressures of agricultural restructuring.
2. Structural shift: In 1989, temporary crops represented 49,6% of total crop holdings, while permanent crops represented 50,4%. By 2019, permanent crops represented 62,2% and temporary crops only 37,8%. Portuguese agriculture therefore shifted from a relatively balanced structure to one increasingly dominated by permanent crops.
3. Policy implications: This shift towards permanent crops may reflect the influence of European Union policies, including Common Agricultural Policy payments and rural development programs, which may have favored permanent crop expansion, particularly olives and vines. These crops are often more profitable per hectare and benefit from European Union subsidies.
4. Environmental implications: The shift to permanent crops may have important environmental implications, including changes in water use through irrigation for olives and vines, soil conservation through permanent cover rather than annual tillage, and biodiversity impacts associated with monoculture expansion.

3.3.4 Concentration and Consolidation

The decline in the number of holdings does not necessarily mean that crop production decreased. Rather, it reflects a profound process of consolidation: fewer, larger farms managing the same or a larger area. However, this consolidation process has not been uniform across agricultural systems. As shown in the previous sections, temporary crops experienced a much steeper decline in the number of holdings (-72,1%) compared with permanent crops (-54,8%). This disparity suggests that the forces associated with consolidation, including mechanization, economies of scale and market pressures, have affected temporary cropping systems more strongly.

The more pronounced consolidation in temporary crops reflects the nature of these production systems. Arable farming, including cereals, pulses and forage crops, is generally more suitable for large-scale mechanized operations, where efficiency gains are achieved by expanding operational size. In contrast, permanent crops such as olive groves and vineyards often require more specialized and labour-intensive management, which may slow down consolidation to some extent. This does not mean that permanent crops have avoided restructuring; rather, their consolidation appears to have followed a different pattern, often involving the modernization and intensification of existing holdings.

These patterns reinforce the broader narrative of agricultural restructuring. Portuguese agriculture is becoming more specialized, more concentrated and more oriented towards permanent crops, while labour and small-scale farming continue to decline. Yet consolidation is not a single, homogeneous process. It is shaped by the specific characteristics of each crop system, by regional conditions and by the policy frameworks that have favored certain types of production over others.

In general, the evidence supports the hypothesis that Portuguese agriculture underwent a process of concentration and consolidation between 1989 and 2019, with temporary crops experiencing the strongest restructuring pressure.

3.4 Permanent grasslands and pastures

Permanent grasslands represent a distinct land-use category in Portuguese agriculture, associated primarily with extensive livestock systems and forage production. Unlike temporary crops or permanent crops, grasslands are not typically cultivated annually and are managed mainly for grazing or hay production. This section examines the evolution of permanent grassland area in mainland Portugal between 1989 and 2019.

Year	Area (ha)	Change from Previous Year (%)	Change from 1989 (%)
1989	754.825	-	-
1999	1.331.033	76,34	76,34
2009	1.721.587	29,34	128,08
2019	2.003.795	16,39	165,46

Table 8: Evolution of Permanent Grassland and Pasture Area (1989 - 2019)

Source: INE – Permanent grasslands area by geographic location (NUTS 2013). Values for "Continente"

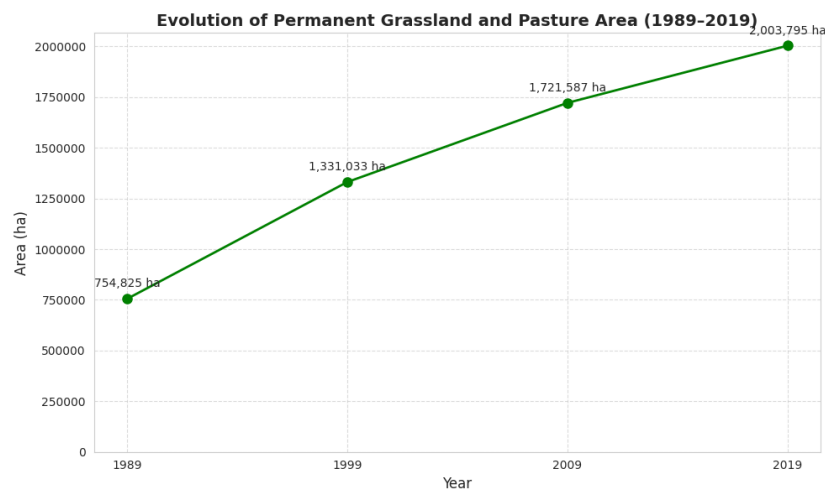


Figure 10: Chart showing the Evolution of Permanent Grassland and Pasture Area

Source: INE – Permanent grasslands area by geographic location (NUTS 2013). Values for "Continente"

The data reveal a striking and consistent increase in permanent grassland area, which more than doubled between 1989 and 2019, growing by 165,46% from 754.825 ha to 2.003.795 ha. This is one of the most pronounced changes among the land-use indicators analyzed.

Three key observations stand out:

- **Accelerated expansion (1989–1999):** Grasslands increased by 576.208 ha (76,34%) in the first decade, the fastest period of expansion. This likely reflects the abandonment of marginal arable land and its conversion to extensive grazing systems, as well as the impact of Common Agricultural Policy measures that supported livestock production.
- **Continued growth (1999–2009):** The expansion continued at a substantial rate, adding another 390.554 ha (29,34%). By 2009, grasslands had become an increasingly important land-use category in many regions, particularly in the interior of the country where arable farming was less viable.
- **Moderating but sustained growth (2009–2019):** While the rate of expansion slowed to 16,39%, grasslands continued to increase, reaching over 2 million hectares by 2019. This suggests a gradual slowdown in the pace of expansion, although the overall trend remains clearly positive.

In general, the evidence supports the interpretation that Portuguese agriculture underwent a significant extensification process in land use, with permanent grasslands expanding alongside broader restructuring trends.

3.4.1 *Permanent Grasslands and the Restructuring of Land Use*

The expansion of permanent grasslands tells a different story from the trends observed in temporary and permanent crops. While crop holdings declined sharply, grasslands expanded significantly. This divergence reflects three important processes:

- **Land use polarization:** Marginal arable land was converted to extensive grazing, while productive land remained in crop production or shifted to permanent crops such as olives and vines. This created a polarized agricultural landscape: intensive cropping in productive areas and extensive livestock in marginal areas.
- **Specialization of livestock systems:** The growth of large grassland holdings suggests a move towards specialized, large-scale livestock operations, consistent with broader trends in European agriculture.
- **Abandonment disguised as extensification:** In many cases, the conversion of arable land to grassland may represent a form of land abandonment, where land remains in agricultural use but with much lower labour and capital inputs. This is consistent with the decline in agricultural labour.

The grassland data therefore adds an important dimension to the restructuring narrative. Portuguese agriculture did not simply decline or intensify uniformly. Instead, it underwent a dual process of intensification in crop production, particularly in permanent crops, and extensification in grassland systems. This dual trajectory is a key characteristic of agricultural change in Mediterranean Europe. The results are consistent with H4, suggesting that permanent grasslands and pastures became an important indicator of land-use transformation, particularly in regions more associated with extensive agricultural systems.

3.5 *Productivity of main agricultural crops*

Understanding how productivity evolved alongside labour and land-use changes is essential for characterizing the restructuring process. If labour declined but productivity increased, this would suggest intensification and efficiency gains. If productivity stagnated or declined alongside labour, it would suggest sectoral decline. This section analyses productivity trends for major crop groups in mainland Portugal between 1989 and 2019.

Rank	Crop Category	Change (%)	Direction
1	Main crops for industry	606,8	↑ Strong growth
2	Main subtropical fruits	183,33	↑ Strong growth
3	Cereals for grain	156,33	↑ Strong growth
4	Olive groves	155,51	↑ Strong growth
5	Citrus fruits	149,78	↑ Strong growth
6	Main fresh fruits	131,20	↑ Strong growth
7	Potatoes	104,77	↑ Strong growth
8	Main dried legumes	42,32	↑ Moderate growth
9	Vineyards	23,61	↑ Moderate growth
10	Main nut fruits	-41,74	↓ Moderate decline
11	Horticultural crops	-	-
12	Main fodder crops	-	-
13	Small berries	-	-

Table 9: Productivity Change Ranking by Crop Category

The data show that productivity increased in most crop categories over the period under analysis. The strongest growth was observed in main crops for industry, followed by main subtropical fruits, cereals for grain, olive groves, citrus fruits, main fresh fruits, potatoes, main dried legumes and vineyards. By contrast, main nut fruits declined, while horticultural crops and main fodder crops do not present comparable values in the available series.

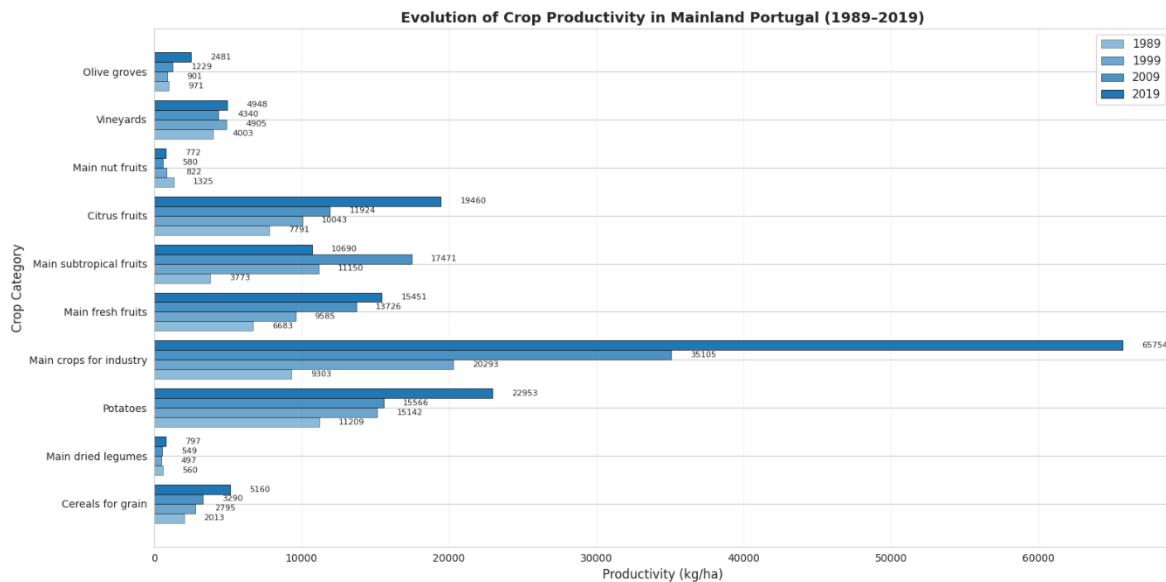


Figure 11: Chart showing the Evolution of crop productivity in mainland Portugal, 1989–2019. Source: Statistics Portugal (INE), crop productivity by geographic location (NUTS 2013), values for “Continente”

These results suggest that agricultural change in mainland Portugal was not limited to labour decline or land-use reallocation. In several crop groups, especially industrial crops, fruit crops and olive groves, productivity gains were very strong, pointing to technological change, improved management and a more market-oriented production structure. At the same time, the more modest growth in vineyards and dried legumes, together with the decline in nut fruits, shows that productivity change was not uniform across all crop systems.

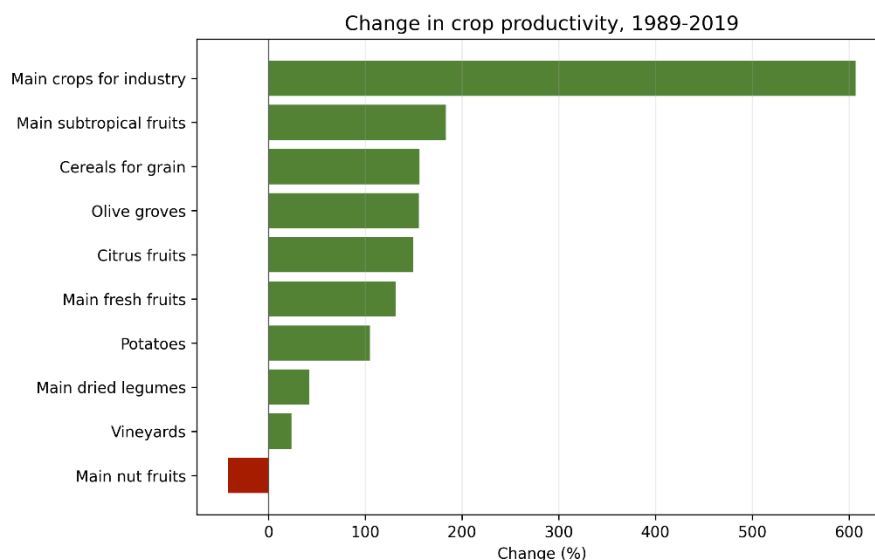


Figure 12: Chart showing Change in crop productivity by crop category, 1989–2019. Source: Statistics Portugal (INE), crop productivity by geographic location (NUTS 2013), values for “Continente”.

Together, the figures show that Portuguese agriculture experienced a selective process of intensification rather than a generalized increase in productivity. This supports the broader interpretation that restructuring involves not only labour decline and land-use change but also differentiated productive outcomes across crop systems. These results support H5, confirming that productivity patterns varied across agricultural activities and that agricultural change was not homogeneous throughout mainland Portugal.

3.5.1 Strong Productivity Growth

Main crops for industry showed the most remarkable increase (+606,8%), from 9.303 to 65.754 kg/ha. This reflects the expansion of intensive, irrigated agriculture in the Alentejo, driven by technological adoption, improved varieties, and investment in irrigation infrastructure.

Main subtropical fruits increased strongly (+183,3%), from 3.773 to 10.690 kg/ha, reflecting the expansion of subtropical fruit production, particularly in the Algarve region, where climatic conditions favor crops like avocados and mangoes.

Cereals for grain increased by +156,3% (from 2.013 to 5.160 kg/ha), consistent with mechanization, improved varieties, and farm consolidation. This aligns with the increase in average temporary holding size.

Olive groves increased by +155,5% (from 971 to 2.481 kg/ha), reflecting the expansion of intensive, irrigated olive production, particularly in the Alentejo region, driven by investment in modern super-high-density orchards.

Citrus fruits increased by +149,8% (from 7.791 to 19.460 kg/ha), reflecting the expansion of citrus production in the Algarve and Ribatejo regions, supported by irrigation investments and improved varieties.

Main fresh fruits increased by +131,2% (from 6.683 to 15.451 kg/ha), reflecting the expansion of intensive fruit orchards in the Oeste and Algarve regions. The adoption of modern cultivation techniques, including trellising and drip irrigation, has contributed significantly to this growth.

Potatoes increased by +104,8% (from 11.209 to 22.953 kg/ha), reflecting concentration in specialized, large-scale operations and the adoption of improved varieties and irrigation.

3.5.2 Moderate Growth

Main dried legumes increased by +42,3% (from 560 to 797 kg/ha), reflecting modest growth in this traditional crop category, though at a much slower pace than the leading sectors.

Vineyards increased by +23,6% (from 4.003 to 4.948 kg/ha), reflecting varietal improvements, investment in trellising and irrigation, and the modernization of Portuguese viticulture in regions like Douro, Dão, and Alentejo.

These categories show that productivity gains were not limited to the most capital-intensive systems, but they were clearly more limited than in industrial crops, citrus or olive groves.

3.5.3 Sharp Decline

Main nut fruits declined by -41,7% (from 1.325 to 772 kg/ha), reflecting challenges in the nut sector, potentially related to market conditions, disease pressures, or competition from other producing regions.

Horticultural crops and Main fodder crops showed N/A values due to data availability limitations, with no consistent time-series data across the study period, preventing trend analysis for these categories.

Small berries reached a productivity of 10.508 kg/ha in 2019, indicating a strong production level in the most recent agricultural census. However, data from earlier years is not available, preventing trend analysis for this category.

4. Inferential Statistics

After identifying the main patterns through exploratory data analysis, this section applies inferential statistics to assess whether the observed trends are statistically meaningful. While the previous section focused on visual interpretation and descriptive comparisons, this stage aims to evaluate the strength and direction of the relationship between time and the selected agricultural indicators.

For this purpose, simple correlation and linear regression tests were applied. The variable **year** was used as the independent variable, while the agricultural indicators were used as dependent variables. This approach makes it possible to examine whether each indicator presents a positive, negative or weak trend over the period analyzed.

	indicator	n	ols_slope_per_decade	slope_ci95_low	slope_ci95_high	pearson_r	ols_p_value	r_squared	spearman_rho	spearman_p_value
0	Agricultural labour	4	-291177.7000	-566926.6839	-15428.7161	-0.9548	0.0452	0.9117	-1.0	0.0
1	Agricultural area	4	-97670.3000	-620109.9152	424769.3152	-0.4944	0.5056	0.2444	-0.8	0.2
2	Holdings with temporary crops	4	-108098.8000	-194327.9985	-21869.6015	-0.9673	0.0327	0.9357	-1.0	0.0
3	Holdings with permanent crops	4	-83937.7000	-161148.7019	-6726.6981	-0.9572	0.0428	0.9162	-1.0	0.0
4	Permanent grasslands and pastures	4	413746.4000	212357.3097	615135.4903	0.9874	0.0126	0.9750	1.0	0.0
5	Agricultural area per worker	4	1.6406	1.2516	2.0295	0.9970	0.0030	0.9940	1.0	0.0

Table 10: **Ordinary least-squares trend test results for agricultural indicators in Mainland Portugal, 1989–2019.**

Note: The table reports slopes per decade with 95% confidence intervals, Pearson and Spearman correlation coefficients, p-values, and R^2 for each indicator. All indicators have $n=4$ observations (census years 1989, 1999, 2009, 2019).

The inferential analysis is organized according to the main hypotheses of the project. Each statistical test is therefore connected to a specific dimension of agricultural transformation, allowing the results to be interpreted in relation to the research question

4.1. Statistical test of agricultural labour (H1)

The linear regression analysis shows a strong negative trend in agricultural labour. The estimated slope was -291,177.7 workers per decade, with a 95% confidence interval ranging from -566,926.7 to -15,428.7. The Pearson correlation coefficient was -0.9548, and the associated p-value of 0.0452 indicates that this downward trend is statistically significant at the 5% level, despite the limited number of observations. A perfect negative Spearman correlation ($\rho = -1.0$, p-value = 0.0) further confirms a strong monotonic decline. Overall, these results support H1 and show that agricultural labour decreased substantially between 1989 and 2019.

4.2. Statistical test of land-use indicators (H2)

The results for total agricultural area show a different pattern. Although the estimated slope was negative (-97,670.3 ha per decade), the 95% confidence interval is wide (-620,109.9 to 424,769.3) and crosses zero, while the p-value is 0.5056. This indicates that there is insufficient evidence to reject the null hypothesis of stability in total area. The weak Pearson correlation (-0.4944) and low R^2 (0.2444) also suggest a weak relationship with time.

However, the descriptive analysis showed that the composition of land use changed significantly, even if total area did not decrease sharply. Therefore, H2 is only partially supported: the overall area does not present a significant linear trend, but the structure of land use clearly changed over time.

4.3. Statistical test of crop systems (H3)

To examine the divergence between temporary and permanent crops, regression tests were performed for both systems. Temporary crops showed a strong and statistically significant declining trend, with a slope of -108,098.8 holdings per decade (p-value = 0.0327; $R^2 = 0.9357$). Permanent crops also showed a significant decline, although less pronounced, with a slope of -83,937.7 holdings per decade (p-value = 0.0428; $R^2 = 0.9162$).

These results support H3. While both systems declined in the number of holdings, the reduction was sharper for temporary crops, indicating a structural shift toward more perennial and potentially more intensive production systems.

4.4. *Statistical test of grasslands and pastures (H4)*

The inferential analysis for permanent grasslands and pastures revealed the strongest and most statistically significant trend in the study. The estimated slope was +413,746.4 hectares per decade. The confidence interval (212,357.3 to 615,135.5) is entirely positive, and the very low p-value (0.0126) confirms statistical significance. The Pearson correlation coefficient (0.9874) and R^2 (0.9750) indicate an almost perfect fit to the linear model. A perfect positive Spearman correlation ($\rho = 1.0$, p-value = 0.0) reinforces this result. These findings strongly support H4 and confirm that the expansion of permanent grasslands is a key feature of agricultural transformation in mainland Portugal.

4.5. *Statistical test of productivity indicators (H5)*

The heterogeneity in productivity trajectories observed in the descriptive analysis is also visible in the regression results. The findings show substantial variability across crop groups. Productivity in crops for industry increased very strongly, and cereals, potatoes, fresh fruits, citrus fruits, and olive groves also recorded notable gains. By contrast, nut fruits declined, and some crop categories could not be analysed due to missing or incomplete time-series values. This variation in productivity patterns supports H5 and shows that agricultural transformation was not uniform. Instead, different cropping systems evolved at different speeds and in different directions, reflecting the uneven nature of structural change in mainland Portuguese agriculture.

5. *Discussion*

The analysis of Portuguese agriculture between 1989 and 2019, based on INE data and a combination of descriptive and inferential statistical methods, reveals a profound and multifaceted restructuring process.

The results show that the sector did not simply decline. Instead, it underwent a deep structural transformation characterized by a sharp reduction in the labour force and the number of agricultural holdings, together with relative stability in total agricultural area. As a result, agricultural areas per worker increased substantially, pointing to greater concentration of production and a stronger role for mechanization.

Land use also changed markedly. The expansion of permanent grasslands and pastures suggests either a strengthening of extensive livestock systems or, in some cases, a form of land abandonment that remained formally within agricultural use. At the same time, the sharp decline in temporary crop holdings compared with permanent crop holdings points to a shift toward more perennial, specialized, and capital-intensive production systems.

Productivity trends reinforce this interpretation of selective restructuring. Industrial crops and several fruit categories recorded strong gains, indicating modernization and technological improvement, while other crops evolved more slowly or even declined. This heterogeneity shows that agricultural change was uneven across production systems and was likely shaped by policy incentives, market conditions, innovation, and regional specificities.

Limitations

6. *Limitations and Future Research Discussion and Conclusion*

The results should be interpreted with caution. The database includes only four census years, which limits the statistical power of the tests. The significance found for indicators such as labour and permanent grasslands should therefore be understood as strong evidence of trend direction, not as definitive proof of causality. Future research could strengthen these findings by using annual data or a longer time series.

The analysis was also conducted at the national level for mainland Portugal, which masks important regional differences. Agricultural dynamics in the Alentejo, for example, are likely very different from those in the Minho, where farm structure and production systems are more fragmented and family-based. Future work should therefore explore regional scales, such as NUTS II or NUTS III, to better understand how these restructuring processes varied across the territory.

7. *Conclusion*

In summary, agriculture in mainland Portugal underwent a major structural transformation between 1989 and 2019. The sector moved away from a more labour-intensive and small-scale model toward a more concentrated, specialized, and capital-dependent system. The sharp decline in agricultural employment, the relative stability of agricultural area, and the strong increase in area per worker are the clearest signs of this change.

The expansion of permanent grasslands and the changing composition of crop systems further show that agricultural restructuring was not uniform. Some activities became more intensive and productive, while others contracted or evolved more slowly. Overall, the evidence indicates that Portuguese agriculture became less labour-dependent and more polarized, with intensive and extensive systems coexisting under the influence of policy, technology, and market forces.

8. Bibliography

Eurostat. (2024). Agriculture statistics. European Commission.
<https://ec.europa.eu/eurostat/web/agriculture>

Instituto Nacional de Estatística. (2020). Recenseamento Agrícola 2019: Análise dos principais resultados. INE.

Instituto Nacional de Estatística. (2021). Recenseamento Agrícola 2019. INE.

Ribeiro, S., & Monteiro, A. (2011). Pastagens permanentes em zonas de montanha: a sua importância e o contributo das medidas agroambientais. HAL Science. Disponível em:
<https://hal.science/hal-01215661/document>

RTP Notícias. (2022). Entre 1989 e 2019. Setor da agricultura perdeu quase um milhão de trabalhadores. https://www.rtp.pt/noticias/economia/entre-1989-e-2019-setor-da-agricultura-perdeu-quase-um-milhao-de-trabalhadores_v1400952

Viver Saudável. (2023). Agricultura portuguesa mudou em 'cinco traços', diz estudo da Fundação Francisco Manuel dos Santos. <https://www.viversaudavel.pt/agricultura-portuguesa-mudou-em-cinco-tracos-diz-estudo-da-fundacao-francisco-manuel-dos-santos/>

Vida Rural. (2025). Produção de vinho deverá cair 11% em 2025. Douro, Lisboa e Alentejo com as maiores quebras. <https://www.vidarural.pt/destaques/producao-de-vinho-portugal/>

Voz do Campo. (2024). Evolução da produção e da produtividade agrícolas nas últimas décadas. <https://vozdocampo.pt/arquivo/29807>

Marques, C. (2012). *As Reformas da Política Agrícola Comum (PAC) e o Sector Agrícola Português*. [https://dspace.uevora.pt/rdpc/bitstream/10174/8532/1/PF%2032-33%20-%20C.%20Marques%20\(135-141\)%2032.4.pdf](https://dspace.uevora.pt/rdpc/bitstream/10174/8532/1/PF%2032-33%20-%20C.%20Marques%20(135-141)%2032.4.pdf)

European Commission. (2026). *Portugal – CAP Strategic Plan*.
https://agriculture.ec.europa.eu/cap-my-country/cap-strategic-plans/portugal_en